

Apple Pascal 1.3 Disk Set: Reconstruction Report

PascalRecon, commit d032f15 2026-09-14. Byte counts are parsed from analysis/diskset-account.txt, written by tools/mkdiskset.py and checked by tools/probes/probe_diskset.py. Finding numbers refer to docs/FINDINGS.md.

Summary

The goal is source for every file on the Apple Pascal 1.3 disk set (128K system) that Apple's own tools, run under AppleWin, turn back into the shipped bytes. The three disks were rebuilt block for block from that work: **359,323 of 430,080 bytes** are identical to Apple's images. Excluding the archived 64K system, **351,873 of 374,784**. Every remaining difference is accounted for by region and has a named cause; the per-region counts add up to a direct compare of the image files.

Category	Bytes	Identical	Differ	What it is
rebuilt	225,792	202,996	22,796	Files produced by Apple's compiler, assembler, Linker, Librarian or SETUP from this repository's source
text	57,344	57,344	0	Text files encoded from src/text/ and their .layout files
boot	3,072	3,069	3	Blocks 0-1 of each disk, from the rebuilt FORMATTER.DATA
directory	6,144	6,076	68	Apple's directory entries re-encoded; dead space written as zero
free	82,432	82,388	44	Unallocated blocks, written as zero
archived	55,296	7,450	47,846	The 64K SYSTEM.APPLE and SYSTEM.PASCAL, out of scope, written as zero
all	430,080	359,323	70,757	The three disks

Of the 35 distinct files: **19 identical**, **13 with a named remainder** (slack, version stamps or session memory that the shipped tools cannot reproduce), **1 accepted exception** (SYSTEM.LIBRARY) and **2 out of scope**.

Files by disk

APPLE1

Blocks	File / region	Bytes	Identical	Differ	Status
0+2	boot blocks	1,024	1,023	1	residue
2+4	directory	2,048	2,034	14	residue
6+32	SYSTEM.APPLE	16,384	2,327	14,057	Out of scope
38+44	SYSTEM.PASCAL	22,528	2,796	19,732	Out of scope
82+50	SYSTEM.EDITOR	25,600	25,600	0	Identical
132+30	SYSTEM.FILER	15,360	15,360	0	Identical
162+38	SYSTEM.LIBRARY	19,456	3,222	16,234	Accepted exception
200+1	SYSTEM.MISCINFO	512	365	147	Named remainder
201+2	SYSTEM.CHARSET	1,024	1,024	0	Identical
203+12	SYSTEM.SYNTAX	6,144	6,144	0	Identical
215+65	free space	33,280	33,280	0	identical
	disk total	143,360	93,175	50,185	

APPLE2

Blocks	File / region	Bytes	Identical	Differ	Status
0+2	boot blocks	1,024	1,023	1	residue
2+4	directory	2,048	2,014	34	residue
6+50	SYSTEM.ASSMBLER	25,600	25,600	0	Identical
56+78	SYSTEM.COMPILER	39,936	39,936	0	Identical
134+25	SYSTEM.LINKER	12,800	12,800	0	Identical
159+8	LIBRARY.CODE	4,096	4,096	0	Identical
167+11	LIBMAP.CODE	5,632	5,321	311	Named remainder
178+2	6502.OPCODES	1,024	1,024	0	Identical
180+7	6502.ERRORS	3,584	3,297	287	Named remainder
187+93	free space	47,616	47,572	44	residue
	disk total	143,360	142,683	677	

APPLE3

Blocks	File / region	Bytes	Identical	Differ	Status
0+2	boot blocks	1,024	1,023	1	residue
2+4	directory	2,048	2,028	20	residue
6+32	SYSTEM.APPLE	16,384	2,327	14,057	Out of scope
38+7	FORMATTER.CODE	3,584	3,190	394	Named remainder
45+7	FORMATTER.DATA	3,584	3,583	1	Named remainder
52+5	BINDER.CODE	2,560	2,544	16	Named remainder
57+2	LINEFEED.CODE	1,024	557	467	Named remainder
59+5	SET40COLS.CODE	2,560	2,544	16	Named remainder
64+1	II40.MISCINFO	512	370	142	Named remainder
65+1	II80.MISCINFO	512	396	116	Named remainder
66+12	BALANCED.TEXT	6,144	6,144	0	Identical
78+8	CROSSREF.TEXT	4,096	4,096	0	Identical
86+22	DISKIO.TEXT	11,264	11,264	0	Identical
108+6	GRAFCHARS.TEXT	3,072	3,072	0	Identical
114+28	GRAFDEMO.TEXT	14,336	14,336	0	Identical
142+1	HAZEL.MISCINFO	512	361	151	Named remainder
143+4	HAZELGOTO.TEXT	2,048	2,048	0	Identical
147+6	HILBERT.TEXT	3,072	3,072	0	Identical
153+33	SETUP.CODE	16,896	12,953	3,943	Named remainder
186+6	SPIRODEMO.TEXT	3,072	3,072	0	Identical
192+8	TREE.TEXT	4,096	4,096	0	Identical
200+32	128K.APPLE	16,384	16,384	0	Identical
232+45	128K.PASCAL	23,040	22,469	571	Named remainder
277+3	free space	1,536	1,536	0	identical
	disk total	143,360	123,465	19,895	

Exceptions and remainders at a glance

File	Differ	Why it cannot be closed
SYSTEM.LIBRARY	16,234	Accepted exception: text blocks from a compiler other than the shipped one (block counts, trailer flag, buffer copies); file two blocks short, so misaligned in place. Unit by unit 16,417 of 19,456 agree, all code identical.
SETUP.CODE	3,943	Apple's file predates the version word and carries 34 stale alignment pads.
128K.PASCAL	571	Slack left by Apple's finishing tool, which is not on the disks.
LINEFEED.CODE	467	1.1 binary: version bits; plus compile slack past the code.
FORMATTER.CODE	394	Linker memory in the last block's slack.
LIBMAP.CODE	311	Linker memory in the last block's slack.
6502.ERRORS	287	The record window's first fill was session memory.
SYSTEM.MISCINFO	147	Memory SETUP never writes (Apple's held SYSTEM.FILER code).
II40.MISCINFO	142	Memory SETUP never writes.
II80.MISCINFO	116	Memory SETUP never writes.
HAZEL.MISCINFO	151	Memory SETUP never writes.
BINDER.CODE	16	1.1 binary: SEGINFO version 2 against a 1.3 compile's 6.
SET40COLS.CODE	16	1.1 binary: SEGINFO version 2 against 6.
FORMATTER.DATA	1	One uncleared assembler buffer byte (the same byte appears in every disk's boot blocks).
SYSTEM.ASSMBLER	0	File identical, but the PASCALIO segment is copied from Apple's file: nothing on the 1.3 disks rebuilds it.
SYSTEM.APPLE	28,114	64K system, archived, not rebuilt (appears on APPLE1 and APPLE3).
SYSTEM.PASCAL	19,732	64K system, archived, not rebuilt.
boot blocks	3	The FORMATTER.DATA byte above, once per disk (3).
directories	68	Dead space past the live entries: Filer text, and on APPLE2 a removed LINKER.INFO entry (68).
free space	44	Block 187 still holds that LINKER.INFO file: linker records for FORMATDI (44).

What the total does not claim: the text files, the boot-block placement and the directory are produced by this repository's encoders and checked against Apple's bytes; no Apple tool wrote them. The directory's names, placement and dates are Apple's history, carried over rather than reconstructed.

Source files by destination

Every file the rebuilt disks carry, and the files it is generated from. Paths are relative to the repository; each one is checked to exist when this report is built. Entries in italics are Apple's shipped files used as inputs, not reconstructed. Each rebuilt file's route through Apple's tools is given in its section under File by file.

Destination	Source files
SYSTEM.COMPILER	src/pascal/1.3/PASCALCO.text — the compiler's declarations and outer procedures src/pascal/1.3/phases/COMPINIT.text — phase segment src/pascal/1.3/phases/COMPOPTI.text — phase segment src/pascal/1.3/phases/DECLARAT.text — phase segment src/pascal/1.3/phases/BODYPART.text — phase segment src/pascal/1.3/phases/ROUTINE.text — phase segment src/pascal/1.3/phases/STATEMEN.text — phase segment src/pascal/1.3/phases/CASESTAT.text — phase segment src/pascal/1.3/phases/FORSTATE.text — phase segment src/pascal/1.3/phases/NUMSTRIN.text — phase segment src/pascal/1.3/phases/BODY1.text — phase segment src/pascal/1.3/phases/BODY3.text — phase segment src/pascal/1.3/phases/UNITPART.text — phase segment src/pascal/1.3/phases/FINISHUP.text — phase segment src/pascal/1.3/phases/WRITELIN.text — phase segment tools/procbuild.py — splices PASCALCO.text and the phases into the one source Apple's compiler reads src/native/SEARCH.TEXT — IDSEARCH and TREESEARCH, 6502, linked in
SYSTEM.ASSMBLER	src/pascal/programs/1.3/ASSMBLER.text — Pascal source, six segments <i>Apple's APPLE2 SYSTEM.ASSMBLER slot 0</i> — the PASCALIO segment, copied by the Librarian, not rebuilt
SYSTEM.LINKER	src/pascal/programs/1.3/LINKER.text — Pascal source
SYSTEM.EDITOR	src/pascal/programs/1.3/EDITOR.text — Pascal source, seven segments
SYSTEM.FILER	src/pascal/programs/1.3/FILER.text — Pascal source
LIBRARY.CODE	src/pascal/programs/1.3/LIBRARY.text — Pascal source (staged as LIBR13.TEXT)
LIBMAP.CODE	src/pascal/programs/1.3/LIBMAP.text — Pascal source src/native/SEARCH.TEXT — IDSEARCH, 6502, linked in
SETUP.CODE	src/pascal/programs/1.3/SETUP.text — Pascal source
BINDER.CODE	src/pascal/programs/1.3/BINDER.text — Pascal source
SET40COLS.CODE	src/pascal/programs/1.3/SET40COLS.text — Pascal source
LINEFEED.CODE	src/pascal/programs/1.3/LINEFEED.text — Pascal source (Apple's own, unaltered)
FORMATTER.CODE	src/pascal/programs/1.3/FORMATTER.text — Pascal source src/native/FORMATTR.TEXT — FORMATDISK, 6502, linked in
FORMATTER.DATA	src/native/ASMFORMAT.TEXT — the Disk II formatter, blocks 0-2 src/native/BOOTII.TEXT — the Disk II boot src/native/BOOTPD.TEXT — the ProDOS blocked-device boot src/pascal/programs/1.3/MAKEBOOT.text — builds BOOTTRACKS.DATA: Disk II boot, blank directory, ProDOS boot src/pascal/programs/1.3/MAKEFMT.text — joins ASMFORMAT.CODE and BOOTTRACKS.DATA into FORMATTER.DATA
128K.APPLE	src/native/interp/INTERP.TEXT — interpreter, bank 2, .ABSOLUTE assembly src/native/interp/TOP.TEXT — interpreter top pages, .ABSOLUTE assembly src/native/interp/BANK1.TEXT — interpreter, bank 1, .ABSOLUTE assembly src/pascal/programs/1.3/MAKEINTP.text — joins the three assemblies' blocks into 128K.APPLE src/native/interp/128K.hints — the trace directives the three .TEXT files are generated from tools/absdis.py — generates the three .TEXT files by tracing Apple's 128K.APPLE with the hints
128K.PASCAL	src/pascal/os/1.3/PASCALSYSTEM.text — Pascal source, seven segments src/pascal/os/1.3/USESFI0.text — include file: the program-level USES FIOPRIMS src/pascal/programs/1.3/MAKEOS.text — finishing program: splits segment 0 as the 128K boot loads it, writes the dictionary

Destination	Source files
SYSTEM.LIBRARY	src/pascal/units/1.3/LONGINTIO.text — unit, slot 0 src/pascal/units/1.3/PASCALIO.text — unit, slot 1 src/pascal/units/1.3/CHAINSTUFF.text — unit, slot 2 src/pascal/units/1.3/TRANSCEND.text — unit, slot 3 src/pascal/units/1.3/TURTLEGRAPHICS.text — unit, slots 4-5 (code and data) src/pascal/units/1.3/APPLESTUFF.text — unit, slot 6 src/pascal/units/1.3/LONGINTIO.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/PASCALIO.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/CHAINSTUFF.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/TRANSCEND.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/TURTLEGRAPHICS.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/APPLESTUFF.layout — how Apple's editor stored the unit's lines src/native/LONGINTS.TEXT — DECOPS, 6502, linked into LONGINTIO src/native/TURTLEGR.TEXT — TURTLEGRAPHICS' native procedures, 6502 src/native/APPLESTF.TEXT — APPLESTUFF's native procedures, 6502
SYSTEM.CHARSET	src/data/CHARSET.TEXT — the character bitmaps src/pascal/programs/1.3/MAKECHRS.text — writes SYSTEM.CHARSET from them
6502.OPCODES	src/data/OPS6502.TEXT — the opcode records src/pascal/programs/1.3/MAKEOPS.text — writes 6502.OPCODES from them
6502.ERRORS	src/data/ERRS6502.TEXT — the error messages src/pascal/programs/1.3/MAKEERRS.text — writes 6502.ERRORS from them
SYSTEM.MISCINFO	src/data/miscinfo/SYSTEM.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into
II40.MISCINFO	src/data/miscinfo/II40.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into
II80.MISCINFO	src/data/miscinfo/II80.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into
HAZEL.MISCINFO	src/data/miscinfo/HAZEL.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into
SYSTEM.SYNTAX	src/text/SYSTEM.SYNTAX.text — the lines src/text/SYSTEM.SYNTAX.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
BALANCED.TEXT	src/text/BALANCED.text — the lines src/text/BALANCED.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
CROSSREF.TEXT	src/text/CROSSREF.text — the lines src/text/CROSSREF.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
DISKIO.TEXT	src/text/DISKIO.text — the lines src/text/DISKIO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
GRAFCHARS.TEXT	src/text/GRAFCHARS.text — the lines src/text/GRAFCHARS.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
GRAFDEMO.TEXT	src/text/GRAFDEMO.text — the lines src/text/GRAFDEMO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
HAZELGOTO.TEXT	src/text/HAZELGOTO.text — the lines src/text/HAZELGOTO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
HILBERT.TEXT	src/text/HILBERT.text — the lines src/text/HILBERT.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
SPIRODEMO.TEXT	src/text/SPIRODEMO.text — the lines src/text/SPIRODEMO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
TREE.TEXT	src/text/TREE.text — the lines src/text/TREE.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)
boot blocks (each disk)	Blocks 3-4 of the rebuilt FORMATTER.DATA, the Disk II boot: src/native/B00TII.TEXT by way of MAKEBOOT and MAKEFMT
directories	Apple's own directory entries, re-encoded by tools/a2pascal/diskwrite.py; the disks are assembled by tools/mkdiskset.py

Destination	Source files
SYSTEM.APPLE, SYSTEM.PASCAL	None: the 64K system is out of scope and not rebuilt.

File by file

SYSTEM.COMPILER [Identical]

Disk	APPLE2	Size	39,936 bytes
Identical	39,936	Differ	0
Findings	90, 105a, 107, 267e		
Source files	src/pascal/1.3/PASCALCO.text — the compiler's declarations and outer procedures src/pascal/1.3/phases/COMPINIT.text — phase segment src/pascal/1.3/phases/COMPOPTI.text — phase segment src/pascal/1.3/phases/DECLARAT.text — phase segment src/pascal/1.3/phases/BODYPART.text — phase segment src/pascal/1.3/phases/ROUTINE.text — phase segment src/pascal/1.3/phases/STATEMEN.text — phase segment src/pascal/1.3/phases/CASESTAT.text — phase segment src/pascal/1.3/phases/FORSTATE.text — phase segment src/pascal/1.3/phases/NUMSTRIN.text — phase segment src/pascal/1.3/phases/BODY1.text — phase segment src/pascal/1.3/phases/BODY3.text — phase segment src/pascal/1.3/phases/UNITPART.text — phase segment src/pascal/1.3/phases/FINISHUP.text — phase segment src/pascal/1.3/phases/WRITELIN.text — phase segment tools/procbuild.py — splices PASCALCO.text and the phases into the one source Apple's compiler reads src/native/SEARCH.TEXT — IDSEARCH and TREESEARCH, 6502, linked in		
Rebuilt by	Apple's compiler, assembler (IDSEARCH/TREESEARCH), Linker, then the Librarian copies slots 1-15 with Apple's notice		
Reconstructed	147 of 147 procedures, 15 segments. A direct descendant of Pascal-P2; names from UCSD's II.0 compiler source where the bytes confirm them.		
Differences	None. All 39,936 bytes, slack included.		

- Slot 0 is blank in Apple's file because the release step was the Librarian, not the compile (finding 105a, closed).

SYSTEM.ASSMBLER [Identical]

Disk	APPLE2	Size	25,600 bytes
Identical	25,600	Differ	0
Findings	235b, 237-266, 267f		
Source files	src/pascal/programs/1.3/ASSMBLER.text — Pascal source, six segments Apple's APPLE2 SYSTEM.ASSMBLER slot 0 — the PASCALIO segment, copied by the Librarian, not rebuilt		
Rebuilt by	Apple's compiler; the Librarian joins slots 1-6 with slot 0 taken from the shipped file		
Reconstructed	90 of 95 procedures in six segments are Apple's compiler's output: TLA 38, ASSEMBLE 33, PROCEND 9, INITIALI 6, SYMTBLDU 3, PRINTERR 1. Names from UCSD's I.5 Adaptable Assembler.		
Differences	None as a file: all 25,600 bytes identical.		

- EXCEPTION: the PASCALIO segment (5 procedures) is copied from Apple's shipped file. It is an older unit (version 2) and nothing on the 1.3 disks can rebuild it (finding 235b). The file is identical, but that segment is borrowed, not reconstructed.

SYSTEM.LINKER [Identical]

Disk	APPLE2	Size	12,800 bytes
Identical	12,800	Differ	0
Findings	268, 291		
Source files	src/pascal/programs/1.3/LINKER.text — Pascal source		
Rebuilt by	Apple's compiler, then the Librarian copies slot 1 with the notice		
Reconstructed	51 of 51 procedures. Ancestor: Roger Sumner's I.5 linker; byte-flip and small-memory names from its II.0 revision.		
Differences	None. All 12,800 bytes.		

- Apple's additions past II.0 (intrinsic units, data segments, version check) keep placeholder names: G29, G91, LK2, LK3, LK17, LK51 and some locals.

SYSTEM.EDITOR [Identical]

Disk	APPLE1	Size	25,600 bytes
Identical	25,600	Differ	0
Findings	272		
Source files	src/pascal/programs/1.3/EDITOR.text — Pascal source, seven segments		
Rebuilt by	Apple's compiler, then the Librarian copies segments 1 and 7-12 into slots 1-7		
Reconstructed	129 of 129 procedures in seven segments, from UCSD's II.0 screen editor.		
Differences	None. All 25,600 bytes.		

SYSTEM.FILER [Identical]

Disk	APPLE1	Size	15,360 bytes
Identical	15,360	Differ	0
Findings	271		
Source files	src/pascal/programs/1.3/FILER.text — Pascal source		
Rebuilt by	Apple's compiler, then the Librarian copies slot 1		
Reconstructed	56 of 56 procedures, one segment, from UCSD's II.0 Filer, inside the 1.3 OS declarations.		
Differences	None. All 15,360 bytes.		

LIBRARY.CODE [Identical]

Disk	APPLE2	Size	4,096 bytes
Identical	4,096	Differ	0
Findings	269, 292		
Source files	src/pascal/programs/1.3/LIBRARY.text — Pascal source (staged as LIBR13.TEXT)		
Rebuilt by	Apple's compiler, then the shipped Librarian copies slot 1		
Reconstructed	16 of 16 procedures. Ancestor: UCSD's I.5 Librarian and its II.0 revision (Dismukes).		
Differences	None. All 4,096 bytes.		

- Apple's own additions keep placeholders: G131, LB6, LB15, M2, M3 and some locals.

128K.APPLE [Identical]

Disk	APPLE3	Size	16,384 bytes
Identical	16,384	Differ	0
Findings	279		
Source files	src/native/interp/INTERP.TEXT — interpreter, bank 2, .ABSOLUTE assembly src/native/interp/TOP.TEXT — interpreter top pages, .ABSOLUTE assembly src/native/interp/BANK1.TEXT — interpreter, bank 1, .ABSOLUTE assembly src/pascal/programs/1.3/MAKEINTP.text — joins the three assemblies' blocks into 128K.APPLE src/native/interp/128K.hints — the trace directives the three .TEXT files are generated from tools/absdis.py — generates the three .TEXT files by tracing Apple's 128K.APPLE with the hints		
Rebuilt by	Apple's assembler (three .ABSOLUTE assemblies), joined by MAKEINTP compiled and run on the 1.3 system		
Reconstructed	The 128K p-machine interpreter.		
Differences	None. All 16,384 bytes.		

- The split between code and data in the three sources is traced, not proven: an absolute assembly has no relocation, so a byte compare cannot tell a data byte written as code from code.

128K.PASCAL [\[Named remainder\]](#)

Disk	APPLE3	Size	23,040 bytes
Identical	22,469	Differ	571
Findings	232-233, 280-284		
Source files	src/pascal/os/1.3/PASCALSYSTEM.text — Pascal source, seven segments src/pascal/os/1.3/USESFI0.text — include file: the program-level USES FIOPRIMS src/pascal/programs/1.3/MAKEOS.text — finishing program: splits segment 0 as the 128K boot loads it, writes the dictionary		
Rebuilt by	Apple's compiler (one compile), finished by MAKEOS compiled and run on the 1.3 system, which splits segment 0 the way the 128K boot loads it		
Reconstructed	All 111 procedures in 7 segments; every segment and SEGKIND Apple's; FIOPRIMS written as INTRINSIC CODE 2.		
Differences	571 bytes: three slack tails that held another tool's memory.		

- Apple's file was finished by a tool that is not on the disks. MAKEOS reproduces the split, the crossing pointers (derived from the 128K boot), word 0x120 and the notice; the slack that tool left cannot be reproduced.

SYSTEM.LIBRARY [\[Accepted exception\]](#)

Disk	APPLE1	Size	19,456 bytes
Identical	3,222	Differ	16,234
Findings	285, 286, 287, 290, 294		
Source files	src/pascal/units/1.3/LONGINTIO.text — unit, slot 0 src/pascal/units/1.3/PASCALIO.text — unit, slot 1 src/pascal/units/1.3/CHAINSTUFF.text — unit, slot 2 src/pascal/units/1.3/TRANSCEND.text — unit, slot 3 src/pascal/units/1.3/TURTLEGRAPHICS.text — unit, slots 4-5 (code and data) src/pascal/units/1.3/APPLESTUFF.text — unit, slot 6 src/pascal/units/1.3/LONGINTIO.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/PASCALIO.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/CHAINSTUFF.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/TRANSCEND.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/TURTLEGRAPHICS.layout — how Apple's editor stored the unit's lines src/pascal/units/1.3/APPLESTUFF.layout — how Apple's editor stored the unit's lines src/native/LONGINTS.TEXT — DECOPS, 6502, linked into LONGINTIO src/native/TURTLEGR.TEXT — TURTLEGRAPHICS' native procedures, 6502 src/native/APPLESTF.TEXT — APPLESTUFF's native procedures, 6502		
Rebuilt by	Apple's compiler (6 units), assembler (3 native halves), Linker (3 links), then the Librarian joins them in Apple's slot order with the notice		
Reconstructed	55 p-code and 14 native procedures. All six code segments and TURTLEGR's data segment byte-identical. Each unit's interface text identical through IMPLEMENTATION.		
Differences	16,417 of 19,456 bytes agree aligned unit by unit. In place on the disk 16,234 differ, because the rebuilt file is two blocks short and every unit after LONGINTI sits one or two blocks early.		

- ACCEPTED EXCEPTION to the whole-file target. Apple's text blocks were written by a compiler other than the shipped one: LONGINTI's text is 2 blocks and TURTLEGR's 3 (the shipped compiler writes 1 and 2), each 10-byte trailer holds N or X at offset 6 (no 1.3 tool writes it), and the extra blocks hold buffer copies.

- Those copies preserve fragments of Apple's own source: LONGINTIO declares PROCEDURE DECOPS; EXTERNAL; first (so the engine was renamed DECOPS), TURTLEGRAPHICS has {\$sendc} and {\$SETC SHORT := FALSE} after IMPLEMENTATION, and a development path /P/NEWC/C.CODE appears in two tails.

- The editor's encoding of each interface is content, so every unit source carries a .layout recording which lines Apple's editor stored with a DLE code and where pages break.

LIBMAP.CODE [\[Named remainder\]](#)

Disk	APPLE2	Size	5,632 bytes
Identical	5,321	Differ	311
Findings	109, 270, 293		
Source files	src/pascal/programs/1.3/LIBMAP.text — Pascal source src/native/SEARCH.TEXT — IDSEARCH, 6502, linked in		
Rebuilt by	Apple's compiler, assembler and Linker (an ordinary program, no Librarian)		
Reconstructed	12 of 12 procedures, native IDSEARCH included; block 0 and all 5,300 bytes before the slack identical. Apple's four added globals named from compiler symbol nodes in the shipped slack.		
Differences	311 bytes of the last block's slack (Linker memory).		

FORMATTER.CODE [\[Named remainder\]](#)

Disk	APPLE3	Size	3,584 bytes
Identical	3,190	Differ	394
Findings	104		
Source files	src/pascal/programs/1.3/FORMATTER.text — Pascal source src/native/FORMATTR.TEXT — FORMATDISK, 6502, linked in		
Rebuilt by	Apple's compiler, assembler and Linker		
Reconstructed	The first whole Pascal-plus-6502 program rebuilt.		
Differences	394 bytes of the last block's slack (Linker memory).		

FORMATTER.DATA [\[Named remainder\]](#)

Disk	APPLE3	Size	3,584 bytes
Identical	3,583	Differ	1
Findings	275		
Source files	src/native/ASMFORMAT.TEXT — the Disk II formatter, blocks 0-2 src/native/BOOTII.TEXT — the Disk II boot src/native/BOOTPD.TEXT — the ProDOS blocked-device boot src/pascal/programs/1.3/MAKEBOOT.text — builds BOOTTRACKS.DATA: Disk II boot, blank directory, ProDOS boot src/pascal/programs/1.3/MAKEFMT.text — joins ASMFORMAT.CODE and BOOTTRACKS.DATA into FORMATTER.DATA		
Rebuilt by	Apple's assembler, joined by two programs compiled and run on the 1.3 system		
Reconstructed	The Disk II formatter, the Disk II boot (also every disk's blocks 0-1) and the ProDOS boot.		
Differences	1 byte: the assembler's uncleared buffer byte, session memory.		

SETUP.CODE [\[Named remainder\]](#)

Disk	APPLE3	Size	16,896 bytes
Identical	12,953	Differ	3,943
Findings	273		
Source files	src/pascal/programs/1.3/SETUP.text — Pascal source		
Rebuilt by	Apple's compiler		
Reconstructed	54 of 54 procedures instruction- and frame-identical, from UCSD's SETUP D1 with Apple's six S.2 edits.		
Differences	3,943 bytes.		

- No 1.3 source can close it: Apple's file predates the SEGINFO version word (0 in every slot) and its compiler left stale bytes in 34 alignment pads that the 1.3 compiler writes as 0.

BINDER.CODE [\[Named remainder\]](#)

Disk	APPLE3	Size	2,560 bytes
Identical	2,544	Differ	16
Findings	99c, 274		
Source files	src/pascal/programs/1.3/BINDER.text — Pascal source		
Rebuilt by	Apple's compiler		
Reconstructed	6 of 6 procedures.		
Differences	16 bytes: SEGINFO version bits.		

- Apple never rebuilt this file for 1.3: it is a 1.1 binary stamped version 2, and any 1.3 compile writes 6. With version 2 written in, the whole file is Apple's.

SET40COLS.CODE [\[Named remainder\]](#)

Disk	APPLE3	Size	2,560 bytes
Identical	2,544	Differ	16
Findings	99c, 112, 274		
Source files	src/pascal/programs/1.3/SET40COLS.text — Pascal source		
Rebuilt by	Apple's compiler		
Reconstructed	4 of 4 procedures.		
Differences	16 bytes: SEGINFO version bits.		

- A 1.1 binary (version 2) carried into 1.3; same cause as BINDER.

LINEFEED.CODE [\[Named remainder\]](#)

Disk	APPLE3	Size	1,024 bytes
Identical	557	Differ	467
Findings	99a, 99c, 288c		
Source files	src/pascal/programs/1.3/LINEFEED.text — Pascal source (Apple's own, unaltered)		
Rebuilt by	Apple's compiler		
Reconstructed	1 procedure, 38 bytes of code.		
Differences	467 bytes: 16 SEGINFO version bits (a 1.1 binary, version 2 against 6) and 451 bytes of compile slack past the code.		

- Every byte through the end of the code is Apple's but the version field.

SYSTEM.CHARSET [\[Identical\]](#)

Disk	APPLE1	Size	1,024 bytes
Identical	1,024	Differ	0
Findings	278		
Source files	src/data/CHARSET.TEXT — the character bitmaps src/pascal/programs/1.3/MAKECHRS.text — writes SYSTEM.CHARSET from them		
Rebuilt by	MAKECHRS compiled and run on the 1.3 system		
Reconstructed	The hi-res character set.		
Differences	None. All 1,024 bytes.		

6502.OPCODES [\[Identical\]](#)

Disk	APPLE2	Size	1,024 bytes
Identical	1,024	Differ	0
Findings	276		
Source files	src/data/OPS6502.TEXT — the opcode records src/pascal/programs/1.3/MAKEOPS.text — writes 6502.OPCODES from them		
Rebuilt by	MAKEOPS compiled and run on the 1.3 system		
Reconstructed	Sixty opcode records for the assembler.		
Differences	None. All 1,024 bytes.		

6502.ERRORS [\[Named remainder\]](#)

Disk	APPLE2	Size	3,584 bytes
Identical	3,297	Differ	287
Findings	276		
Source files	src/data/ERRS6502.TEXT — the error messages src/pascal/programs/1.3/MAKEERRS.text — writes 6502.ERRORS from them		
Rebuilt by	MAKEERRS compiled and run on the 1.3 system		
Reconstructed	85 STRING[40] records, record 65 rewritten afterwards as Apple's was.		
Differences	287 bytes, all descended from the record window's first fill (session memory). Apple's fill substituted into ours gives Apple's file.		

SYSTEM.MISCINFO [\[Named remainder\]](#)

Disk	APPLE1	Size	512 bytes
Identical	365	Differ	147
Findings	277		
Source files	src/data/miscinfo/SYSTEM.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into		
Rebuilt by	Apple's shipped SETUP, driven from the recipe on the 1.3 system		
Reconstructed	Screen and keyboard profile for the boot system: every setting and the zero tail.		
Differences	The rest of the 512 bytes is memory SETUP never writes.		

- Apple's SYSTEM.MISCINFO held SYSTEM.FILER's code there; a 128K run does not leave it.

II40.MISCINFO [\[Named remainder\]](#)

Disk	APPLE3	Size	512 bytes
Identical	370	Differ	142
Findings	277		
Source files	src/data/miscinfo/II40.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into		
Rebuilt by	Apple's shipped SETUP, driven from the recipe on the 1.3 system		
Reconstructed	40-column Apple II profile: every setting and the zero tail.		
Differences	The rest of the 512 bytes is memory SETUP never writes.		

II80.MISCINFO [\[Named remainder\]](#)

Disk	APPLE3	Size	512 bytes
Identical	396	Differ	116
Findings	277		
Source files	src/data/miscinfo/II80.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into		
Rebuilt by	Apple's shipped SETUP, driven from the recipe on the 1.3 system		
Reconstructed	80-column profile: every setting and the zero tail.		
Differences	The rest of the 512 bytes is memory SETUP never writes.		

HAZEL.MISCINFO [\[Named remainder\]](#)

Disk	APPLE3	Size	512 bytes
Identical	361	Differ	151
Findings	277		
Source files	src/data/miscinfo/HAZEL.recipe — every SETUP field's value Apple's APPLE3 SETUP.CODE — Apple's shipped SETUP, which the recipe is typed into		
Rebuilt by	Apple's shipped SETUP, driven from the recipe on the 1.3 system		
Reconstructed	Hazeltine terminal profile: every setting and the zero tail.		
Differences	The rest of the 512 bytes is memory SETUP never writes.		

SYSTEM.SYNTAX [\[Identical\]](#)

Disk	APPLE1	Size	6,144 bytes
Identical	6,144	Differ	0
Findings	288		
Source files	src/text/SYSTEM.SYNTAX.text — the lines src/text/SYSTEM.SYNTAX.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.
- Plain text with a DLE code on six lines only (the other nine files have one on nearly every line).

BALANCED.TEXT [\[Identical\]](#)

Disk	APPLE3	Size	6,144 bytes
Identical	6,144	Differ	0
Findings	288		
Source files	src/text/BALANCED.text — the lines src/text/BALANCED.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

CROSSREF.TEXT [\[Identical\]](#)

Disk	APPLE3	Size	4,096 bytes
Identical	4,096	Differ	0
Findings	288		
Source files	src/text/CROSSREF.text — the lines src/text/CROSSREF.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

DISKIO.TEXT [Identical]

Disk	APPLE3	Size	11,264 bytes
Identical	11,264	Differ	0
Findings	288		
Source files	src/text/DISKIO.text — the lines src/text/DISKIO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

GRAFCHARS.TEXT [Identical]

Disk	APPLE3	Size	3,072 bytes
Identical	3,072	Differ	0
Findings	288		
Source files	src/text/GRAFCHARS.text — the lines src/text/GRAFCHARS.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

GRAFDEMO.TEXT [Identical]

Disk	APPLE3	Size	14,336 bytes
Identical	14,336	Differ	0
Findings	288		
Source files	src/text/GRAFDEMO.text — the lines src/text/GRAFDEMO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

HAZELGOTO.TEXT [Identical]

Disk	APPLE3	Size	2,048 bytes
Identical	2,048	Differ	0
Findings	288		
Source files	src/text/HAZELGOTO.text — the lines src/text/HAZELGOTO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.
- Its page zero is not SYSTEM.EDITOR's HEADER record: the environment sits 142 bytes later; recorded as raw bytes.

HILBERT.TEXT [Identical]

Disk	APPLE3	Size	3,072 bytes
Identical	3,072	Differ	0
Findings	288		
Source files	src/text/HILBERT.text — the lines src/text/HILBERT.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

SPIRODEMO.TEXT [Identical]

Disk	APPLE3	Size	3,072 bytes
Identical	3,072	Differ	0
Findings	288		
Source files	src/text/SPIRODEMO.text — the lines src/text/SPIRODEMO.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

TREE.TEXT [Identical]

Disk	APPLE3	Size	4,096 bytes
Identical	4,096	Differ	0
Findings	288		
Source files	src/text/TREE.text — the lines src/text/TREE.layout — DLE codes, page breaks and the editor's page zero tools/a2pascal/textfile.py — the encoder (Layout)		
Rebuilt by	Encoded by tools/a2pascal/textfile.py (Layout)		
Reconstructed	Lines, per-line DLE encoding, greedy page packing, and the editor's page zero (margins, command character, created and last-used dates).		
Differences	None.		

- Produced by this repository's encoder from the text and its .layout, checked against Apple's bytes; no Apple tool is run.

SYSTEM.APPLE [Out of scope]

Disk	APPLE1, APPLE3 (figures are for each copy)	Size	16,384 bytes
Identical	2,327	Differ	14,057
Findings	--		
Source files	None: not rebuilt.		
Rebuilt by	Not rebuilt		
Reconstructed	The 64K system. Out of scope by decision; the build writes zero here.		
Differences	Every nonzero byte of Apple's counts as a difference.		

- Archived with Apple Pascal 1.1; the project targets the 128K system only.

SYSTEM.PASCAL [Out of scope]

Disk	APPLE1	Size	22,528 bytes
Identical	2,796	Differ	19,732
Findings	--		
Source files	None: not rebuilt.		
Rebuilt by	Not rebuilt		
Reconstructed	The 64K system. Out of scope by decision; the build writes zero here.		
Differences	Every nonzero byte of Apple's counts as a difference.		

- Archived with Apple Pascal 1.1; the project targets the 128K system only.

How it was checked

Acceptance tier. Apple's own compiler, assembler, Linker, Librarian and SETUP, run under AppleWin on the 128K 1.3 system and driven over a redirected console (tools/emuremote.py). Their output is kept verbatim in acceptance/, and probes compare it with Apple's disks on every build (python tools/build_all.py, which must exit 0).

Fast tier. Host reimplementations (ucsdpsys_compile, tools/asm6502.py) can only falsify, never accept.

The disk account. tools/mkdiskset.py writes APPLE1-3 from the rebuilt files. probe_diskset.py requires the regions to tile each disk, their difference counts to equal a direct compare of the image files, the same build from Apple's own regions to reproduce Apple's images exactly (so placement is not a source of difference), and each region to differ by exactly the count its finding explains.

Names. Identifiers are taken from UCSD's I.5 and II.0 sources, Apple's own source fragments and compiler symbol nodes left in slack, and only where the code matches. Apple's own additions with no ancestor keep placeholder names (LK<n>, G<n>, L<n>); renames were re-verified byte-identical on the acceptance tier (findings 291-294).

Labels. Every claim in docs/FINDINGS.md is labelled VERIFIED BINARY FACT, VERIFIED SOURCE FACT, STRONG INFERENCE or SPECULATION.